

MATH 1075 - Section 7.5 Worksheet

09/22/2026

Simplifying a Complex Fraction (Method I)

Step 1 Add or subtract expressions in the numerator to form a single fraction. Add or subtract expressions in the denominator to form a single fraction.

Step 2 Divide the rational expressions from step 1 by multiplying the numerator of the complex fraction by the reciprocal of the denominator of the complex fraction.

Step 3 Simplify to lowest terms if possible.

Simplify following complex fractions using method 1

1. $\frac{\frac{a^2}{2a-3}}{\frac{5a}{8a-12}}$

2. $\frac{\frac{12x^2}{5y}}{\frac{8x^6}{9y^2}}$

3. $\frac{\frac{8}{9}-\frac{1}{3}}{\frac{7}{6}+\frac{1}{9}}$

4. $\frac{\frac{1}{b}+1}{\frac{1}{b}}$

5. $\frac{2+\frac{1}{x}}{4+\frac{1}{x}}$

Simplifying a Complex Fraction (Method II)

Step 1 Multiply the numerator and denominator of the complex fraction by the LCD of *all* individual fractions within the expression.

Step 2 Apply the distributive property, and simplify the numerator and denominator.

Step 3 Simplify to lowest terms if possible.

Simplify following complex fractions using method 2

1. $\frac{2 + \frac{1}{x}}{4 + \frac{1}{x}}$

2. $\frac{\frac{8}{9} - \frac{1}{3}}{\frac{7}{6} + \frac{1}{9}}$

3. $\frac{1 - \frac{9}{p^2}}{1 - \frac{1}{p} - \frac{6}{p^2}}$

4. $\frac{\frac{9}{4m} + \frac{9}{2m^2}}{\frac{3}{2} + \frac{3}{m}}$